EN.553.634 Elements of Statistical Learning

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0.1 Introduction to Statistical Learning

Let $(X, Y) \sim P$, where $X \in \mathcal{X} \subseteq \mathbb{R}^p$ is a feature vector and Y is a response. Statistical learning uses a sample $\mathcal{D}_n = \{(X_i, Y_i)\}_{i=1}^n$ to construct a prediction rule $f : \mathcal{X} \to \mathcal{A}$. In regression, $\mathcal{A} = \mathbb{R}$; in classification, $\mathcal{A}$ is a finite set or a vector of class probabilities.

Prediction and interpretation are different goals. A predictive model is judged on new observations from the deployment distribution. A coefficient can be interpreted causally only with additional identification assumptions; predictive accuracy alone does not provide them.

Example 0.1.1 (From coin flips to supervised learning)**.** Suppose a coin's probability of heads depends on measurable conditions X, and $Y \in \{0, 1\}$ records the outcome. Estimating one bias p becomes estimating the conditional probability

$$\eta(x) = P(Y = 1 \mid X = x).$$

A classifier converts $\eta(x)$ into an action, while a calibrated probability model attempts to estimate $\eta(x)$ itself.

0.1.1 Loss Functions

A loss $L(y, a)$ quantifies the cost of predicting a when the outcome is y. Common choices are

$$L_{\text{sq}}(y,a) = (y-a)^2, \qquad L_{\text{abs}}(y,a) = |y-a|,$$
$$L_{01}(y,a) = \mathbb{1}\{y \neq a\}, \qquad L_{\log}(y,q) = -y\log q - (1-y)\log(1-q).$$

Squared loss strongly penalizes large residuals; absolute loss is more robust. Zero-one loss targets classification accuracy but is discontinuous and hard to optimize. Logistic loss is a smooth convex surrogate and estimates class probabilities when the model is well specified.

A loss should reflect the decision. For asymmetric binary costs c_{10} for a false positive and c_{01} for a false negative,

$$L(y,a) = c_{10}\mathbb{1}\{y=0, a=1\} + c_{01}\mathbb{1}\{y=1, a=0\}.$$

0.1.2 Risk and Empirical Risk Minimization

The population and empirical risks are

$$R(f) = \mathbb{E}[L(Y, f(X))], \qquad \widehat{R}_n(f) = \frac{1}{n}\sum_{i=1}^{n} L(Y_i, f(X_i)). \tag{0.1.1}$$

Given a function class $\mathcal{F}$, empirical risk minimization (ERM) chooses $\widehat{f} \in \arg\min_{f\in\mathcal{F}} \widehat{R}_n(f)$. If f^* minimizes risk over all measurable rules, then

$$R(\widehat{f}) - R(f^*) = \underbrace{R(\widehat{f}) - \inf_{f\in\mathcal{F}} R(f)}_{\text{estimation error}} + \underbrace{\inf_{f\in\mathcal{F}} R(f) - R(f^*)}_{\text{approximation error}}.$$

The first term reflects finite data; the second reflects the restricted model class. Interpolation of training data alone gives no guarantee of generalization.

Write $\widehat{f}_{\mathcal{D}}$ when emphasizing dependence on the training sample. Population risk $R(g)$ evaluates a fixed rule g, whereas $R(\widehat{f}_{\mathcal{D}})$ is random before $\mathcal{D}$ is drawn.

Example 0.1.2 (The optimal constant rule)**.** Without predictors and under squared loss,

$$R(a) = \mathbb{E}[(Y-a)^2] = \text{Var}(Y) + \{\mathbb{E}(Y) - a\}^2, \qquad \widehat{R}_{\mathcal{D}}(a) = \frac{1}{n}\sum_{i=1}^{n}(Y_i - a)^2.$$

The Bayes rule is $a^* = \mu = \mathbb{E}Y$, the ERM is $\widehat{a}_{\mathcal{D}} = \bar{Y}$, and the minimum population risk is $\text{Var}(Y)$. For $\widehat{\mu}_\alpha = \alpha\bar{Y}$,

$$\text{MSE}(\widehat{\mu}_\alpha) = \frac{\alpha^2\sigma^2}{n} + (\alpha-1)^2\mu^2.$$

Differentiation gives the oracle shrinkage $\alpha^* = \mu^2/(\mu^2 + \sigma^2/n)$. Although unavailable without knowing μ and σ^2, it demonstrates how bias can reduce MSE.

More explicitly,

$$\mathrm{Bias}_{\mathcal{D}}(\widehat{\mu}_\alpha) = \mathbb{E}_{\mathcal{D}}[\widehat{\mu}_\alpha] - \mu = (\alpha - 1)\mu, \qquad \mathrm{Var}_{\mathcal{D}}(\widehat{\mu}_\alpha) = \frac{\alpha^2\sigma^2}{n},$$

so the preceding MSE is variance plus squared bias. Substitution of α^* gives the minimum oracle MSE

$$\mathrm{MSE}(\widehat{\mu}_{\alpha^*}) = \frac{\mu^2\sigma^2}{n\mu^2 + \sigma^2}.$$

Now let Y_0 be a new response, independent of the training sample and drawn from the same population. Conditional on the fitted value $\widehat{\mu}_\alpha$, its generalization error is

$$\begin{aligned} R_{\mathcal{D}}(\widehat{\mu}_\alpha) &= \mathbb{E}\Big[(Y_0 - \widehat{\mu}_\alpha)^2 \mid \mathcal{D}\Big] \\ &= \sigma^2 + (\mu - \widehat{\mu}_\alpha)^2. \end{aligned}$$

Averaging over repeated training samples yields

$$\mathbb{E}_{\mathcal{D}} R_{\mathcal{D}}(\widehat{\mu}_\alpha) = \underbrace{\sigma^2}_{\text{irreducible error}} + \underbrace{\mathrm{MSE}(\widehat{\mu}_\alpha)}_{\text{estimation error}}.$$

Thus estimation MSE and expected prediction error are not identical: predicting a fresh response adds its irreducible variance σ^2.

0.1.3 Optimal Prediction

Conditioning on $X = x$ reduces prediction to a pointwise decision:

$$f^*(x) \in \arg\min_a \mathbb{E}[L(Y, a) \mid X = x]. \tag{0.1.2}$$

Under squared loss, $f^*(x) = \mathbb{E}[Y \mid X = x]$; under absolute loss, any conditional median is optimal. Under zero-one loss, the optimal classifier chooses the most probable class. More precisely, if $Y \in \{1, \ldots, K\}$ and $\eta_k(x) = P(Y = k \mid X = x)$, then

$$g^*(x) \in \arg\max_{1 \le k \le K} \eta_k(x), \qquad \min_a \mathbb{E}[\mathbb{1}\{Y \neq a\} \mid X = x] = 1 - \max_{1 \le k \le K} \eta_k(x).$$

Thus regression and classification share the same conditional-risk framework; changing the response space and loss changes the resulting Bayes rule.

Squared-loss calculation. Writing $m(x) = \mathbb{E}[Y \mid X = x]$,

$$\mathbb{E}[(Y - a)^2 \mid X = x] = \mathrm{Var}(Y \mid X = x) + \{m(x) - a\}^2,$$

which is minimized by $a = m(x)$. □

0.1.4 The Regression Function

The regression function $m(x) = \mathbb{E}[Y \mid X = x]$ describes the systematic part of a response. Writing

$$Y = m(X) + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0,$$

does not require normality or constant variance. The conditional variance $\sigma^2(x) = \mathrm{Var}(Y \mid X = x)$ may depend on x. For Bernoulli Y, $m(x) = P(Y = 1 \mid X = x)$ is both the regression function and the class probability.

For any rule g, squared-loss risk decomposes as

$$R(g) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathbb{E}[\{m(X) - g(X)\}^2]. \tag{0.1.3}$$

The first term is irreducible Bayes error; the second is reducible error.

0.1.5 Nearest-Neighbor Regression and the Curse of Dimensionality

For continuous X, exact replicates at $X = x$ are unlikely. The k-nearest-neighbor estimator averages locally:

$$\widehat{f}_{k,\mathcal{D}}(x) = \frac{1}{k} \sum_{i:X_i \in N_k(x)} Y_i.$$

Small k tends toward low smoothing bias and high variance; large k lowers variance at the price of bias. Scaling determines the meaning of distance.

To capture fraction r of a uniform distribution in the unit p-cube, a smaller cube needs side length

$$\ell = r^{1/p}. \tag{0.1.4}$$

For $r = 0.1$, ℓ is approximately 0.32 in dimension two, 0.63 in dimension five, and 0.79 in dimension ten. A neighborhood containing enough observations is no longer local. Structured model classes overcome this curse by sharing information across feature space, but introduce approximation bias.

The same phenomenon appears in a Euclidean ball. If X is uniform in the unit ball in $\mathbb{R}^p$, scaling of volume gives

$$P(\|X\|_2 \le r) = r^p, \qquad 0 \le r \le 1.$$

Consequently, its median radius is $2^{-1/p}$, which approaches 1 as $p \to \infty$. Thus half the volume lies outside a radius increasingly close to the boundary. In high dimensions, even a neighborhood containing a moderate fraction of the population must extend almost across the entire support.

0.1.6 Structured Function Classes

When the data are too sparse for reliable local averaging, learning requires assumptions that share information across input values. For a scalar feature, a degree-q polynomial class is

$$\mathcal{F}_q = \left\{ x \mapsto \beta_0 + \beta_1 x + \cdots + \beta_q x^q : (\beta_0, \ldots, \beta_q) \in \mathbb{R}^{q+1} \right\}.$$

These classes are nested:

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots .$$

Therefore their best empirical risks satisfy

$$\inf_{f \in \mathcal{F}_{q+1}} \widehat{R}_n(f) \leq \inf_{f \in \mathcal{F}_q} \widehat{R}_n(f).$$

Enlarging a model class can never worsen its optimized training fit, although it may worsen generalization error.

Linear and polynomial models are parametric: a finite coefficient vector fixes their functional form. Nearest-neighbor averaging is local and nonparametric: its effective complexity grows with the sample and it does not impose one global formula. More structure typically improves stability and interpretability but can increase approximation bias; less structure offers greater flexibility at the price of higher variance and weaker interpretability.

0.1.7 Generalization Error

Conditional on the training data, test error is

$$\mathrm{Err}(\mathcal{D}_n) = \mathbb{E}[L(Y^{\mathrm{new}}, \widehat{f}(X^{\mathrm{new}})) \mid \mathcal{D}_n].$$

Training error reuses observations that selected $\widehat{f}$ and is optimistically biased. Test observations must be excluded from every modeling choice, including imputation, scaling, feature selection, and tuning.

Ordinary validation assumes that training and deployment observations have the same distribution. Covariate shift changes P_X but not $P_{Y|X}$; concept drift changes the conditional law. Random cross-validation on historical data does not diagnose either automatically.

For an independent test pair (X_0, Y_0), the precise target is

$$R(\widehat{f}_{\mathcal{D}}) = \mathbb{E}[(Y_0 - \widehat{f}_{\mathcal{D}}(X_0))^2 \mid \mathcal{D}]. \tag{0.1.5}$$

Training MSE reuses the observations that selected the rule and is typically optimistic. For nested classes, increasing flexibility cannot increase training error, but test error can increase through greater sampling variability.

0.1.8 The Bias-Variance Tradeoff

For squared-error prediction at fixed x, averaging over training sets gives

$$\mathbb{E}[(Y_0 - \widehat{f}(x))^2 \mid X_0 = x] = \sigma^2(x) + \{\mathbb{E}[\widehat{f}(x)] - m(x)\}^2 + \mathrm{Var}(\widehat{f}(x)). \tag{0.1.6}$$

These are irreducible noise, squared bias, and variance. Restrictive models tend toward higher bias and lower variance; flexible models tend toward lower approximation bias and higher sampling variance. This exact additive decomposition is specific to squared loss.

With training-set randomness shown explicitly,

$$\mathrm{Bias}_{\mathcal{D}}\{\widehat{f}_{\mathcal{D}}(x)\} = \mathbb{E}_{\mathcal{D}}[\widehat{f}_{\mathcal{D}}(x)] - m(x).$$

Averaging over a new predictor gives

$$\begin{aligned}\mathbb{E}_{\mathcal{D}} R(\widehat{f}_{\mathcal{D}}) = \mathbb{E}_X[\mathrm{Var}(Y \mid X)] + \mathbb{E}_X[\mathrm{Bias}_{\mathcal{D}}\{\widehat{f}_{\mathcal{D}}(X)\}^2] \\ + \mathbb{E}_X[\mathrm{Var}_{\mathcal{D}}\{\widehat{f}_{\mathcal{D}}(X)\}].\end{aligned}$$

Training error is not a term in this decomposition and can fall below Bayes error, which bounds new-observation risk rather than error on reused data.

0.2 Least Squares

Given $X \in \mathbb{R}^{n\times p}$ and $y \in \mathbb{R}^n$, ordinary least squares solves

$$\widehat{\beta} \in \operatorname*{arg\,min}_{\beta\in\mathbb{R}^p} \|y - X\beta\|^2 . \tag{0.2.1}$$

The normal equations are $X^\mathsf{T}X\widehat{\beta} = X^\mathsf{T}y$. If X has full column rank,

$$\widehat{\beta} = (X^\mathsf{T}X)^{-1}X^\mathsf{T}y.$$

Computationally, solve the system using QR or SVD rather than explicitly forming the inverse; forming the Gram matrix squares the condition number.

The derivation follows by expanding

$$\mathrm{RSS}(\beta) = y^\mathsf{T}y - 2\beta^\mathsf{T}X^\mathsf{T}y + \beta^\mathsf{T}X^\mathsf{T}X\beta.$$

Therefore

$$\nabla_\beta \mathrm{RSS}(\beta) = -2X^\mathsf{T}y + 2X^\mathsf{T}X\beta, \qquad \nabla^2_\beta \mathrm{RSS} = 2X^\mathsf{T}X \succeq 0.$$

Setting the gradient to zero gives the normal equations, and positive semidefiniteness shows that every solution is a global minimizer.

The best population linear predictor and its ERM estimate are distinct:

$$\beta^* = \operatorname*{arg\,min}_\beta \mathbb{E}[(Y - X^\mathsf{T}\beta)^2], \qquad \widehat{\beta}_{\mathcal{D}} = \operatorname*{arg\,min}_\beta \frac{1}{n}\sum_{i=1}^n (Y_i - X_i^\mathsf{T}\beta)^2.$$

Even when $m(x)$ is nonlinear, $x^\mathsf{T}\beta^*$ is the best linear approximation under the population distribution.

To derive the population coefficient, expand

$$\mathbb{E}[(Y - X^\mathsf{T}\beta)^2] = \mathbb{E}[Y^2] - 2\beta^\mathsf{T}\mathbb{E}[XY] + \beta^\mathsf{T}\mathbb{E}[XX^\mathsf{T}]\beta.$$

Differentiating gives the population normal equations

$$\mathbb{E}[XX^\mathsf{T}]\beta^* = \mathbb{E}[XY].$$

If $\mathbb{E}[XX^\mathsf{T}]$ is invertible, then

$$\beta^* = \mathbb{E}[XX^\mathsf{T}]^{-1}\mathbb{E}[XY]. \tag{0.2.2}$$

In contrast, the empirical normal equations divide by n to give

$$\left(\frac{1}{n}X^\mathsf{T}X\right)\widehat{\beta} = \frac{1}{n}X^\mathsf{T}y,$$

and hence

$$\widehat{\beta} = \left(\frac{1}{n}X^\mathsf{T}X\right)^{-1}\left(\frac{1}{n}X^\mathsf{T}y\right).$$

Thus OLS is the plug-in analogue of the best population linear predictor: the empirical second moment and cross-moment replace their population counterparts.

0.2.1 Estimation and Geometry

The fitted vector is the orthogonal projection of y onto the column space:

$$\widehat{y} = Hy, \qquad H = X(X^\mathsf{T}X)^{-1}X^\mathsf{T}.$$

The hat matrix is symmetric and idempotent. The residual $e = y - \widehat{y}$ is orthogonal to every column of X. If X is rank deficient, fitted values remain unique but coefficients do not; X^+y is the minimum-norm solution.

Leverage $h_{ii} = H_{ii}$ measures how unusual row i is in feature space. Since $\sum_i h_{ii} = \operatorname{tr}(H) = p$, average leverage is p/n. The leave-one-out residual satisfies

$$y_i - \widehat{y}_i^{(-i)} = \frac{e_i}{1 - h_{ii}}, \tag{0.2.3}$$

so OLS leave-one-out error can be computed without n refits.

The normal equations are equivalently $X^\mathsf{T}e = 0$, so $e \perp \operatorname{Col}(X)$ and $y = \widehat{y} + e$ is an orthogonal decomposition. Adding a column to X enlarges the projection space; therefore RSS and training error cannot increase.

0.2.2 Sampling Properties of Least Squares

Under the fixed-design model

$$y = X\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0, \qquad \operatorname{Cov}(\varepsilon \mid X) = \sigma^2 I,$$

OLS is conditionally unbiased and

$$\operatorname{Cov}(\widehat{\beta} \mid X) = \sigma^2(X^\mathsf{T}X)^{-1}.$$

The derivation is

$$\begin{aligned}\widehat{\beta} &= (X^\mathsf{T}X)^{-1}X^\mathsf{T}(X\beta + \varepsilon)\\ &= \beta + (X^\mathsf{T}X)^{-1}X^\mathsf{T}\varepsilon,\end{aligned}$$

so conditional expectation gives unbiasedness and

$$\begin{aligned}\operatorname{Cov}(\widehat{\beta} \mid X) &= (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}(\sigma^2 I)X(X^{\mathsf{T}}X)^{-1} \\ &= \sigma^2(X^{\mathsf{T}}X)^{-1}.\end{aligned}$$

For simple regression with an intercept,

$$\operatorname{Var}(\widehat{\beta}_1 \mid X) = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}.$$

The slope is estimated more precisely with lower noise, more observations, or greater predictor spread. Gauss–Markov says OLS has smallest covariance among linear unbiased estimators; it does not require Gaussian errors. Normality gives exact t and F inference. An unbiased error-variance estimator is

$$\widehat{\sigma}^2 = \frac{\left\| y - X\widehat{\beta} \right\|^2}{n - p}.$$

With heteroskedastic errors, use sandwich standard errors or weighted least squares. If $\mathbb{E}[\varepsilon \mid X] \neq 0$, robust standard errors do not remove coefficient bias.

Example 0.2.1 (Omitted-variable bias)**.** If $Y = \beta_0 + \beta_1 X + \beta_2 Z + \varepsilon$ but Z is omitted, the population slope from regressing Y on X is

$$\beta_1 + \beta_2 \frac{\operatorname{Cov}(X, Z)}{\operatorname{Var}(X)}.$$

Bias requires both a nonzero effect of Z and association between X and Z.

0.2.3 Confidence and Prediction Intervals

Let d be the number of fitted columns of X. Under conditionally Gaussian errors,

$$\frac{\widehat{\beta}_j - \beta_j}{\widehat{\sigma}\sqrt{[(X^{\mathsf{T}}X)^{-1}]_{jj}}} \sim t_{n-d}.$$

Thus a $(1 - \alpha)$ confidence interval for β_j is

$$\widehat{\beta}_j \pm t_{n-d,1-\alpha/2}\widehat{\sigma}\sqrt{[(X^{\mathsf{T}}X)^{-1}]_{jj}}.$$

Across repetitions of the sampling procedure, the stated fraction of such intervals cover the fixed coefficient.

At a new feature vector x_0,

$$\operatorname{Var}\{x_0^{\mathsf{T}}\widehat{\beta} \mid X\} = \sigma^2 x_0^{\mathsf{T}}(X^{\mathsf{T}}X)^{-1}x_0.$$

A confidence interval for the mean response $x_0^{\mathsf{T}}\beta$ uses this variance. A prediction interval for $Y_0 = x_0^{\mathsf{T}}\beta + \varepsilon_0$ uses

$$\widehat{\sigma}\sqrt{1 + x_0^{\mathsf{T}}(X^{\mathsf{T}}X)^{-1}x_0}. \tag{0.2.4}$$

It is wider because it includes irreducible noise in the new response.

0.2.4 Generalization Error and the Cost of Dimension

At fixed x_0, unbiasedness gives

$$\mathbb{E}_{\mathcal{D},Y_0}[(Y_0 - x_0^\mathsf{T}\widehat{\beta})^2 \mid X] = \sigma^2 + \sigma^2 x_0^\mathsf{T}(X^\mathsf{T}X)^{-1}x_0.$$

Let $\Sigma_X = \mathbb{E}[X_0X_0^\mathsf{T}]$ and suppose $n^{-1}X^\mathsf{T}X \approx \Sigma_X$. Averaging over X_0 gives

$$\mathbb{E}_{X_0}[X_0^\mathsf{T}(X^\mathsf{T}X)^{-1}X_0] \approx \frac{1}{n}\operatorname{tr}(\Sigma_X^{-1}\Sigma_X) = \frac{d}{n}.$$

Consequently,

$$\mathbb{E}[\text{generalization error}] \approx \sigma^2\left(1 + \frac{d}{n}\right), \qquad \mathbb{E}[\text{training error}] = \sigma^2\left(1 - \frac{d}{n}\right). \tag{0.2.5}$$

The expected optimism is approximately $2\sigma^2 d/n$. Under correct linear specification, added dimensions cost estimation variance; under misspecification, they may also reduce approximation bias.

For a fixed design, the corresponding training-error calculation is exact. Let

$$H = X(X^\mathsf{T}X)^{-1}X^\mathsf{T}, \qquad e = (I - H)y.$$

Under the correctly specified model $y = X\beta + \varepsilon$, the identity $(I-H)X = 0$ gives $e = (I-H)\varepsilon$. Since $I - H$ is symmetric and idempotent,

$$\begin{aligned}\mathbb{E}[\|e\|_2^2 \mid X] &= \mathbb{E}[\varepsilon^\mathsf{T}(I - H)\varepsilon \mid X] \\ &= \sigma^2 \operatorname{tr}(I - H) = \sigma^2(n - d).\end{aligned}$$

Therefore

$$\mathbb{E}\left[\frac{1}{n}\|y - X\widehat{\beta}\|_2^2 \middle| X\right] = \sigma^2\left(1 - \frac{d}{n}\right).$$

There is also an exact fixed-design version of the test-error calculation. At row x_i^T of X, a new independent response has expected squared prediction error

$$\sigma^2 + \sigma^2 x_i^\mathsf{T}(X^\mathsf{T}X)^{-1}x_i = \sigma^2(1 + h_{ii}).$$

Averaging over the n observed predictor rows and using $\sum_i h_{ii} = \operatorname{tr}(H) = d$ yields

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[(Y_i^{\text{new}} - x_i^\mathsf{T}\widehat{\beta})^2 \mid X] = \sigma^2\left(1 + \frac{d}{n}\right).$$

The gap between new-response and training error is consequently exactly $2\sigma^2 d/n$ for this fixed-design comparison.

0.3 Multiple Regression

An intercept, numeric predictors, and encoded categorical variables are columns of a design matrix. With an intercept, a K-level categorical predictor uses $K-1$ reference indicators. The coefficient of X_j is the change in fitted response per unit change in X_j, holding the other represented features fixed.

The Frisch–Waugh–Lovell theorem says the coefficient of X_j equals the slope obtained by residualizing both Y and X_j against the remaining predictors. Multiple regression therefore compares variation in X_j not linearly explained by the other columns.

To see this algebraically, partition the design as $X = [x_1\ X_2]$ and let

$$P_2 = X_2(X_2^\mathsf{T}X_2)^{-1}X_2^\mathsf{T}, \qquad M_2 = I - P_2.$$

For any fixed β_1, minimizing over the coefficients of X_2 projects $y - x_1\beta_1$ onto $\mathrm{Col}(X_2)$. The remaining objective is therefore

$$\|M_2(y - x_1\beta_1)\|_2^2.$$

Differentiating with respect to β_1 gives

$$\widehat{\beta}_1 = \frac{x_1^\mathsf{T}M_2y}{x_1^\mathsf{T}M_2x_1} = \frac{(M_2x_1)^\mathsf{T}(M_2y)}{(M_2x_1)^\mathsf{T}(M_2x_1)}. \tag{0.3.1}$$

Thus $\widehat{\beta}_1$ is the simple-regression slope between the part of x_1 unexplained by X_2 and the part of y unexplained by X_2. If x_1 is orthogonal to every column of X_2, then $M_2x_1 = x_1$, so adding X_2 does not change its coefficient.

0.3.1 Interactions and Basis Expansions

An interaction permits one effect to depend on another:

$$m(x_1, x_2) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_{12}x_1x_2.$$

Then $\partial m/\partial x_1 = \beta_1 + \beta_{12}x_2$, so β_1 is the effect of x_1 only at $x_2 = 0$. Centering can improve interpretation.

For a binary group indicator $Z \in \{0, 1\}$, the additive model

$$m(x, z) = \beta_0 + \beta_1x + \beta_2z$$

fits parallel lines: group 0 has intercept β_0, group 1 has intercept $\beta_0 + \beta_2$, and both have slope β_1. Adding an interaction,

$$m(x, z) = \beta_0 + \beta_1x + \beta_2z + \beta_3xz,$$

allows group 1 to have slope $\beta_1 + \beta_3$. Hence β_2 is the between-group intercept change at $x = 0$, while β_3 is the between-group slope change.

A basis expansion uses fixed transformations h_m:

$$f(x) = \sum_{m=1}^{M} \beta_m h_m(x).$$

The model is nonlinear in x but linear in β. Raw high-degree polynomials are often unstable; orthogonal polynomials or splines are usually better conditioned.

0.4 Logistic Regression

For $Y \in \{0, 1\}$, logistic regression models

$$p(x) = P(Y = 1 \mid X = x), \qquad \log \frac{p(x)}{1 - p(x)} = \beta_0 + x^{\mathsf{T}}\beta. \tag{0.4.1}$$

Thus $p(x) = \{1 + \exp[-(\beta_0 + x^{\mathsf{T}}\beta)]\}^{-1}$. A unit increase in x_j multiplies conditional odds by e^{β_j} holding other modeled features fixed; it does not add a constant amount to probability.

0.4.1 Log Loss and Probability Estimation

When the prediction is a probability $q \in (0, 1)$ rather than a class label, the binary log loss is

$$L_{\log}(y, q) = -y \log q - (1 - y) \log(1 - q).$$

Fix x and write $p = P(Y = 1 \mid X = x)$. The conditional risk of reporting q is

$$R_x(q) = -p \log q - (1 - p) \log(1 - q).$$

Its derivatives are

$$R'_x(q) = -\frac{p}{q} + \frac{1 - p}{1 - q} = \frac{q - p}{q(1 - q)}, \qquad R''_x(q) = \frac{p}{q^2} + \frac{1 - p}{(1 - q)^2} > 0.$$

Hence the unique minimizer is

$$q^*(x) = p = P(Y = 1 \mid X = x).$$

Thus zero–one loss targets the most likely class, whereas log loss targets the entire conditional probability. Equivalently,

$$R_x(q) = H(p) + D_{\mathrm{KL}}\Big(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(q)\Big),$$

so any incorrect probability incurs nonnegative excess conditional risk.

0.4.2 Likelihood

Including the intercept in x_i, the Bernoulli log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} [y_i x_i^\mathsf{T} \beta - \log\{1 + e^{x_i^\mathsf{T}\beta}\}]. \tag{0.4.2}$$

The negative log-likelihood is exactly the empirical log loss:

$$-\ell(\beta) = \sum_{i=1}^{n} L_{\log}\Big(y_i, p_\beta(x_i)\Big).$$

Consequently, logistic maximum likelihood is empirical risk minimization under log loss. If $p_i = p(x_i)$, then

$$\nabla\ell(\beta) = X^\mathsf{T}(y - p), \qquad \nabla^2\ell(\beta) = -X^\mathsf{T} W X,$$

where $W = \mathrm{diag}\{p_i(1 - p_i)\}$. The likelihood is concave.

Under perfect separation, a hyperplane correctly separates all observations and the likelihood supremum occurs as $\|\beta\| \to \infty$; no finite unpenalized MLE exists. Ridge regularization gives a finite solution. Numerically, evaluate $\log(1 + e^z)$ with a stable softplus routine.

0.4.3 Fitting Logistic Regression: Newton and IRLS

Newton's update is

$$\beta^{\text{new}} = \beta^{\text{old}} + (X^\mathsf{T} W X)^{-1} X^\mathsf{T}(y - p). \tag{0.4.3}$$

With working response $z = X\beta + W^{-1}(y - p)$, this is weighted least squares:

$$\beta^{\text{new}} = (X^\mathsf{T} W X)^{-1} X^\mathsf{T} W z.$$

This is iteratively reweighted least squares (IRLS). Implementations solve the weighted system rather than invert it, and use step halving or line search if a full Newton step decreases the likelihood. Growing coefficients and fitted probabilities near zero or one suggest separation.

0.5 Bayes Classification

Let $Y \in \{1, \ldots, K\}$ and $\eta_k(x) = P(Y = k \mid X = x)$. Under zero-one loss, the Bayes classifier is

$$g^*(x) = \arg\max_k \eta_k(x),$$

with Bayes error $R^* = \mathbb{E}[1 - \max_k \eta_k(X)]$.

0.5.1 Plug-in Classification and kNN

A plug-in classifier first estimates the conditional class probabilities and then substitutes them into the Bayes rule:

$$\widehat{g}(x) \in \arg\max_k \widehat{\eta}_k(x).$$

For k-nearest-neighbor classification, let $N_k(x)$ denote the indices of the k training inputs closest to x. Estimate

$$\widehat{\eta}_c(x) = \frac{1}{k} \sum_{i \in N_k(x)} \mathbb{1}\{Y_i = c\}, \qquad c = 1, \ldots, K,$$

and predict the local majority class. Small k produces a flexible boundary with low smoothing bias and high sampling variance; large k produces a smoother, less variable boundary with greater bias.

For direct multiclass probability modeling, softmax regression uses

$$p_k(x) = \frac{\exp(\alpha_k + x^\mathsf{T}\beta_k)}{\sum_{j=1}^{K} \exp(\alpha_j + x^\mathsf{T}\beta_j)}, \qquad k = 1, \ldots, K.$$

The probabilities are positive and sum to one. After fitting by multiclass log loss, the corresponding zero–one-loss prediction is $\widehat{g}(x) = \arg\max_k \widehat{p}_k(x)$. One class or one coefficient vector must be fixed as a reference because adding the same score to every class does not change the probabilities.

Generative classifiers model priors π_k and class densities f_k, then apply

$$P(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_j \pi_j f_j(x)}.$$

0.5.2 Linear Discriminant Analysis

Linear discriminant analysis assumes

$$X \mid Y = k \sim \mathcal{N}(\mu_k, \Sigma), \qquad k = 1, \ldots, K,$$

with a common covariance matrix across classes. For classification, the denominator in Bayes' theorem is common to all k, so it suffices to maximize $\log \pi_k + \log f_k(x)$. Apart from constants common to every class,

$$\begin{aligned} \log \pi_k + \log f_k(x) &= -\frac{1}{2}(x - \mu_k)^\mathsf{T}\Sigma^{-1}(x - \mu_k) + \log \pi_k + \text{constant} \\ &= x^\mathsf{T}\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\mathsf{T}\Sigma^{-1}\mu_k + \log \pi_k - \frac{1}{2}x^\mathsf{T}\Sigma^{-1}x + extconstant. \end{aligned}$$

The final quadratic term does not depend on k and cancels when classes are compared. Define the LDA discriminant score

$$\delta_k(x) = x^\mathsf{T}\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\mathsf{T}\Sigma^{-1}\mu_k + log\pi_k. \tag{0.5.1}$$

Then

$$g_{\mathrm{LDA}}(x) = \arg\max_k \delta_k(x).$$

Since each δ_k is affine in x, the pairwise boundary $\delta_k(x) = \delta_\ell(x)$ is a hyperplane. The class means, covariance, and priors determine its orientation and location.

Given n_k observations in class k, the usual fitted quantities are

$$\widehat{\pi}_k = \frac{n_k}{n}, \qquad \widehat{\mu}_k = \frac{1}{n_k} \sum_{i:Y_i=k} X_i,$$

and the pooled within-class covariance

$$\widehat{\Sigma} = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:Y_i=k} (X_i - \widehat{\mu}_k)(X_i - \widehat{\mu}_k)^\mathsf{T}.$$

Substituting these estimates into (0.5.1) produces the fitted LDA rule.

0.5.3 Quadratic Discriminant Analysis

Quadratic discriminant analysis instead assumes

$$X \mid Y = k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

Its discriminant score is

$$\delta_k^{\mathrm{QDA}}(x) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2}(x-\mu_k)^\mathsf{T} \Sigma_k^{-1} (x-\mu_k) + \log \pi_k. \tag{0.5.2}$$

Because the covariance differs by class, the terms $x^\mathsf{T}\Sigma_k^{-1}x$ no longer cancel. Pairwise decision boundaries are therefore quadratic.

LDA estimates one covariance matrix with $p(p+1)/2$ free entries, whereas QDA estimates K such matrices. QDA can reduce bias when class covariances truly differ, but its greater parameter count increases variance and requires more data. LDA imposes more structure and is typically more stable.

For two classes, the posterior log-odds make the relationship to logistic regression explicit:

$$\log \frac{P(Y=1 \mid X=x)}{P(Y=0 \mid X=x)} = \log \frac{\pi_1 f_1(x)}{\pi_0 f_0(x)}.$$

Shared covariance makes this expression affine in x; distinct covariances make it quadratic. Logistic regression models the posterior log-odds directly, whereas LDA and QDA obtain them from a model for $X \mid Y$.

0.5.4 Naive Bayes

Naive Bayes assumes the feature coordinates are conditionally independent within each class:

$$f_k(x) = p(X = x \mid Y = k) = \prod_{j=1}^{p} f_{kj}(x_j).$$

The resulting discriminant score can be written

$$\log \pi_k + \sum_{j=1}^{p} \log f_{kj}(x_j).$$

Each marginal f_{kj} may be modeled according to its feature type, for example Gaussian, Bernoulli, Poisson, or nonparametrically. Gaussian naive Bayes corresponds to class-specific diagonal covariance matrices. It uses far fewer covariance parameters than QDA, at the cost of the strong conditional independence assumption.

0.5.5 Multiclass Decisions under a Loss Matrix

Estimating class probabilities and choosing an action are distinct steps. Let $L(j,k)$ be the loss for predicting class j when the true class is k. At a fixed x, the conditional risk of action j is

$$R_x(j) = \sum_{k=1}^{K} L(j,k)\eta_k(x), \qquad \eta_k(x) = P(Y = k \mid X = x).$$

The Bayes action is

$$g_L^*(x) \in \arg\min_{1\le j\le K} R_x(j). \tag{0.5.3}$$

Only under zero–one loss does this reduce to choosing the most probable class.

For example, suppose

$$\eta(x) = \begin{pmatrix} 0.45 \\ 0.35 \\ 0.20 \end{pmatrix}, \qquad L = \begin{pmatrix} 0 & 1 & 5 \\ 1 & 0 & 1 \\ 5 & 1 & 0 \end{pmatrix},$$

where rows are predictions and columns are true classes. Then

$$R_x(1) = 1.35, \qquad R_x(2) = 0.65, \qquad R_x(3) = 2.60.$$

Class 1 is most probable, but class 2 minimizes expected loss because it avoids the costly confusion between classes 1 and 3.

0.5.6 Thresholds

With false-positive cost c_{10} and false-negative cost c_{01}, predict class 1 when

$$\eta(x) > \frac{c_{10}}{c_{10} + c_{01}}. \tag{0.5.4}$$

Indeed, fixing x and writing $\eta(x) = P(Y = 1 \mid X = x)$, the two conditional risks are

$$R_x(1) = c_{10}\{1 - \eta(x)\}, \qquad R_x(0) = c_{01}\eta(x).$$

The inequality $R_x(1) < R_x(0)$ rearranges to $\eta(x) > c_{10}/(c_{10} + c_{01})$, which proves Eq. (0.5.4). Thus a threshold is not an arbitrary convention: it encodes the relative consequences of the two errors. The threshold $1/2$ is optimal only for equal costs. Changing a threshold changes decisions, not ranking or calibration. Select it on validation data using deployment costs or capacity constraints.

For a threshold rule $g_t(x) = \mathbb{1}\{\widehat{\eta}(x) > t\}$, let $\pi_1 = P(Y = 1)$ and $\pi_0 = P(Y = 0)$. Its population cost can be written in terms of its ROC operating point:

$$\begin{aligned} R(g_t) &= c_{01} P(g_t(X) = 0, Y = 1) + c_{10} P(g_t(X) = 1, Y = 0) \\ &= c_{01}\pi_1\{1 - \mathrm{TPR}(t)\} + c_{10}\pi_0 \,\mathrm{FPR}(t). \end{aligned} \tag{0.5.5}$$

Thus the preferred ROC operating point depends on error costs and class prevalence; the curve alone does not select a deployment threshold.

0.5.7 ROC and Calibration

For binary classification, the confusion matrix records true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). The principal rates are

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad \mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}, \qquad \text{specificity} = 1 - \mathrm{FPR}\,.$$

TPR is also called sensitivity or recall. These are conditional rates and should not be confused with precision $\mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$, which depends directly on class prevalence.

For threshold t,

$$\mathrm{TPR}(t) = P(\widehat{\eta}(X) > t \mid Y = 1), \qquad \mathrm{FPR}(t) = P(\widehat{\eta}(X) > t \mid Y = 0).$$

The ROC curve plots TPR against FPR. AUC is the probability that a random positive receives a higher score than a random negative, with half credit for ties. It assesses ranking, not probability accuracy, and may hide behavior in the operational low-FPR region.

To see the ranking interpretation, let S^+ and S^- be independent scores drawn from the positive and negative classes, and let F_- be the cdf of S^-. For continuous scores,

$$\begin{aligned} \mathrm{AUC} &= \int \mathrm{TPR}(t)\, dF_-(t) \\ &= \int P(S^+ > t)\, dF_-(t) = \mathbb{E}\{P(S^+ > S^- \mid S^-)\} \\ &= P(S^+ > S^-). \end{aligned}$$

This also shows why any strictly increasing transformation of the scores leaves the ROC curve and AUC unchanged. When scores are discrete, ties may have positive probability and the precise pairwise formula is

$$\mathrm{AUC} = P(S^+ > S^-) + \frac{1}{2} P(S^+ = S^-).$$

Consequently a score independent of the class has AUC 1/2, whether the score distribution is continuous or discrete.

For multiclass evaluation, one-versus-rest calculations produce a metric for each class. A *macro average* gives every class equal weight, for example

$$\mathrm{TPR}_{\mathrm{macro}} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{TPR}_k\,.$$

A *micro average* first pools all one-versus-rest counts:

$$\mathrm{TPR}_{\mathrm{micro}} = \frac{\sum_k \mathrm{TP}_k}{\sum_k (\mathrm{TP}_k + \mathrm{FN}_k)}.$$

Macro averaging exposes performance on rare classes, while micro averaging is dominated by common classes.

A score is calibrated when $P(Y = 1 \mid \widehat{\eta}(X) = q) = q$. Reliability plots, log loss, and Brier score assess calibration. Platt scaling fits a logistic recalibration; isotonic regression fits a monotone map. Under class imbalance, precision–recall curves are often more informative because precision depends on prevalence.

0.6 Estimating Generalization Error

The quantity of interest is performance of the entire training procedure, including tuning. A test set estimates this only if it is untouched until the procedure is finalized. Repeatedly checking test performance turns the test set into training information.

Let $D = \{(X_i, Y_i)\}_{i=1}^n$ and let $\widehat{f}_D$ denote the predictor produced by a learning procedure. Its training error is

$$\widehat{R}_D(\widehat{f}_D) = \frac{1}{n} \sum_{i=1}^{n} L\{Y_i, \widehat{f}_D(X_i)\},$$

whereas, conditional on the realized training set, its generalization error is

$$R(\widehat{f}_D) = \mathbb{E}\Big[L\{Y, \widehat{f}_D(X)\} \mid D \Big], \tag{0.6.1}$$

for a fresh (X, Y) from the population. Training error is generally optimistic because the same observations were used both to choose $\widehat{f}_D$ and to assess it.

An independent test set T gives the conditional estimate

$$\widehat{R}_T(\widehat{f}_D) = \frac{1}{|T|} \sum_{(X_i, Y_i) \in T} L\{Y_i, \widehat{f}_D(X_i)\}.$$

Once test performance is used to choose among procedures, however, the chosen index $\widehat{m} = \arg\min_m \widehat{R}_T(\widehat{f}_D^{(m)})$ depends on T. Consequently $\widehat{R}_T(\widehat{f}_D^{(\widehat{m})})$ is optimistic as a final assessment. The three distinct roles are therefore: training data fit parameters, validation data select the procedure, and untouched test data assess the finalized procedure.

0.6.1 Validation and CV

In a validation split, train on one subset, choose hyperparameters on another, and evaluate once on a test subset. In K-fold cross-validation (CV), partition indices into folds $I_1, \ldots, I_K$ and compute

$$\widehat{\mathrm{Err}}_{\mathrm{CV}} = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} L\{Y_i, \widehat{f}^{(-k)}(X_i)\}.$$

Five- or ten-fold CV is a common variance–computation compromise. Leave-one- out CV has nearly maximal training sets but can have high variance and is not always preferable.

More explicitly, leave-one-out cross-validation (LOOCV) fits $\widehat{f}_{D_{-i}}$ on $D_{-i} = D \setminus \{(X_i, Y_i)\}$ and reports

$$\widehat{R}_{\mathrm{LOO}} = \frac{1}{n} \sum_{i=1}^{n} L\{Y_i, \widehat{f}_{D_{-i}}(X_i)\}. \tag{0.6.2}$$

Each response is assessed by a predictor that did not use it. Thus LOOCV targets the expected generalization error of the *learning procedure* trained on $n - 1$ observations, rather than the conditional error of one fixed fitted model. The distinction from training on all n observations is often small for stable procedures and large n, but it is conceptually important. For least squares, substituting Eq. (0.2.3) into squared-error loss gives

$$\widehat{R}_{\mathrm{LOO}} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}} \right)^2,$$

so one full-data fit suffices.

Preprocessing must be refit within each training fold. For grouped, longitudinal, spatial, or time-ordered data, folds must respect dependence and deployment direction. Random row splits can leak subjects or future information.

"Preprocessing" includes every data-dependent operation: feature screening, standardization, imputation, transformations, dimensionality reduction, and model fitting. For example, selecting the predictors most correlated with Y using the full dataset before CV leaks validation responses into the training process. The selection must instead be repeated separately inside each training fold.

Nested CV uses inner folds for tuning and outer folds for evaluation. The one-standard-error rule chooses the simplest model whose estimated error is within one standard error of the minimum, favoring stability when differences are mostly noise.

0.6.2 Bootstrap and Statistical Uncertainty

The nonparametric bootstrap samples n training observations with replacement, refits the complete procedure, and approximates sampling variability from the replicates. About $1 - e^{-1} \approx 0.632$ of distinct observations appear in a large bootstrap sample; the rest are out-of-bag observations.

Formally, if $D = \{Z_1, \ldots, Z_n\}$ consists of i.i.d. observations from an unknown distribution P, the empirical distribution is

$$\widehat{P}_D = \frac{1}{n} \sum_{i=1}^{n} \delta_{Z_i}, \qquad \widehat{P}_D(Z = Z_i) = \frac{1}{n}.$$

The bootstrap substitutes $\widehat{P}_D$ for P and draws conditionally independent datasets

$$D^{*(1)}, \ldots, D^{*(B)} \overset{\text{iid}}{\sim} \widehat{P}_D^{,n}.$$

Recomputing a statistic T on each sample gives the bootstrap distribution $T^{*(1)}, \ldots, T^{*(B)}$, which approximates the repeated-sampling distribution of $T(D)$. This is a conditional simulation: the observed data are held fixed while the bootstrap randomness varies.

The 0.632 fact follows by fixing observation Z_i. It is absent on one draw with probability $1 - 1/n$, hence

$$P_*(Z_i \notin D^*) = \left(1 - \frac{1}{n}\right)^n \longrightarrow e^{-1}, \qquad P_*(Z_i \in D^*) \longrightarrow 1 - e^{-1}.$$

Bootstrap standard errors estimate variability of parameters or predictions:

$$\widehat{\mathrm{se}}_{\mathrm{boot}}(\widehat{\theta}) = \left\{ \frac{1}{B-1} \sum_{b=1}^{B} (\widehat{\theta}^{*(b)} - \bar{\theta}^*)^2 \right\}^{1/2}.$$

Percentile, basic, and BCa intervals make different corrections. The ordinary bootstrap can fail for nonsmooth statistics, parameters on boundaries, extreme tails, or dependent observations. Use block bootstrap for suitable dependent sequences.

What is resampled must match the intended sampling experiment. Resampling observed pairs (X_i, Y_i) lets both the design and response vary and targets random-design uncertainty. If regression inference is conditional on fixed X, a residual bootstrap instead fits the model, centers the residuals $e_i = Y_i - \widehat{Y}_i$, samples e_i^* with replacement, constructs

$$Y_i^* = \widehat{Y}_i + e_i^*,$$

and refits using the original X_i. Independently resampling observations from a time series destroys serial dependence; a suitable block bootstrap preserves local dependence. In all cases, the bootstrap should imitate the population experiment whose uncertainty is being estimated.

0.6.2.1 Bootstrap Aggregation

Bootstrap replicates can also be averaged rather than merely inspected for their spread. For predictions, bootstrap aggregation (*bagging*) uses

$$\widehat{f}_{\mathrm{bag}}(x) = \frac{1}{B} \sum_{b=1}^{B} \widehat{f}_{D^{*(b)}}(x) \longrightarrow \mathbb{E}_*\{\widehat{f}_{D^*}(x) \mid D\}.$$

This can reduce variance for unstable methods, whose fitted rules change substantially after small perturbations of the training data.

The effect is especially transparent for 1-nearest-neighbor regression. Fix x_0 and order the training points by distance as $X_{(1)}, \ldots, X_{(n)}$. Let J be the rank of the first point that appears in a size-n bootstrap sample. Then

$$w_j = P_*(J = j) = \left(\frac{n-j+1}{n}\right)^n - \left(\frac{n-j}{n}\right)^n \longrightarrow e^{-(j-1)} - e^{-j}.$$

Therefore infinitely bagged 1-NN is the weighted-neighbor rule

$$\mathbb{E}_*\{\widehat{f}^{\mathrm{1NN}}_{D^*}(x_0) \mid D\} = \sum_{j=1}^{n} w_j Y_{(j)} \approx .632Y_{(1)} + .233Y_{(2)} + .086Y_{(3)} + \cdots .$$

Bagging has changed the learning rule from a discontinuous single-neighbor prediction into a smoother average of nearby responses.

CV fold errors are correlated because fitted training sets overlap; treating them as independent observations understates uncertainty. Comparing two models by paired out-of-fold losses is better than comparing separate averages, but repeated or nested resampling is needed for honest selection uncertainty.

0.7 Model Selection

Model selection chooses features, transformations, hyperparameters, or an algorithm. Training fit always improves as a nested model grows, so selection must penalize complexity or estimate out-of-sample error.

For Gaussian linear models, common criteria are

$$\mathrm{AIC} = -2\ell(\widehat{\theta}) + 2d, \qquad \mathrm{BIC} = -2\ell(\widehat{\theta}) + d\log n,$$

where d is the number of fitted parameters. AIC targets predictive Kullback–Leibler risk; BIC more strongly favors small models and, under restrictive conditions, consistently selects a true finite-dimensional model. They answer different questions.

Searching many models creates selection bias. Standard errors and p-values computed as if the selected model had been fixed in advance are generally too optimistic.

0.7.1 Subset Selection

For a subset $S \subseteq \{1, \ldots, p\}$, define the least-squares training criterion

$$\mathrm{RSS}(S) = \min_{\beta_0, \{\beta_j : j \in S\}} \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j \in S} \beta_j X_{ij} \right)^2 .$$

Best-subset selection retains, for every size k, the candidate

$$S_k^* \in \arg\min_{|S|=k} \mathrm{RSS}(S).$$

There are $\binom{p}{k}$ candidates of size k and 2^p subsets in total, so exhaustive search quickly becomes infeasible. Although $\mathrm{RSS}(S_k^*)$ cannot increase with k, the best subsets need not be nested: S_k^* need not be contained in S_{k+1}^*. Training RSS is used to find the best candidate *within* each size; CV or a complexity criterion must still choose the size.

Forward stepwise selection starts with $S_0 = \varnothing$ and greedily updates

$$j_k^* \in \arg\min_{j \notin S_k} \mathrm{RSS}(S_k \cup \{j\}), \qquad S_{k+1} = S_k \cup \{j_k^*\}.$$

Backward stepwise selection starts with all predictors and greedily removes

$$j_k^* \in \arg\min_{j \in S_k} \mathrm{RSS}(S_k \setminus \{j\}), \qquad S_{k-1} = S_k \setminus \{j_k^*\}.$$

Both produce nested paths after roughly $1 + p(p+1)/2$ fits, but their greedy paths need not contain the best subset of a given size. Backward selection also requires that the full model be fit, so it is unavailable in ordinary least squares when $p \geq n$. When CV evaluates either procedure, the entire selection path must be recomputed inside each training fold.

0.7.2 Complexity and Degree of Freedom

Degrees of freedom measure how strongly fitted values respond to the observed responses, not simply the number of nonzero parameters.

For a fixed-design Gaussian model $Y = \mu + \varepsilon$ with $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$, the covariance definition is

$$\mathrm{df}(\widehat{Y}) = \frac{1}{\sigma^2} \sum_{i=1}^{n} \mathrm{Cov}(\widehat{Y}_i, Y_i \mid X). \tag{0.7.1}$$

If $\widehat{Y} = HY$ is a linear smoother, then

$$\mathrm{Cov}(\widehat{Y}_i, Y_i \mid X) = \sum_j H_{ij} \,\mathrm{Cov}(Y_j, Y_i \mid X) = \sigma^2 H_{ii},$$

and hence $\mathrm{df}(\widehat{Y}) = \mathrm{tr}(H)$. For OLS with a fixed, rank-d design,

$$\mathrm{tr}\{X(X^\mathsf{T}X)^{-1}X^\mathsf{T}\} = \mathrm{tr}\{(X^\mathsf{T}X)^{-1}X^\mathsf{T}X\} = d.$$

Adaptive search consumes additional degrees of freedom beyond the number of coefficients eventually retained. For a simple example, let $Y_1, Y_2 \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and fit whichever coordinate has larger absolute value:

$$\widehat{Y} = \begin{cases} (Y_1, 0), & |Y_1| \geq |Y_2|, \\ (0, Y_2), & |Y_2| > |Y_1|. \end{cases}$$

Only one coordinate is retained, yet Eq. (0.7.1) gives

$$\mathrm{df}(\widehat{Y}) = \frac{1}{\sigma^2}\mathbb{E}\{\max(Y_1^2, Y_2^2)\} = 1 + \frac{2}{\pi} \approx 1.64 > 1.$$

The excess is the cost of using Y to search for the apparently stronger coordinate.

Mallows' C_p and Stein's unbiased risk estimate correct training error by an optimism term proportional to degrees of freedom. Complexity can also be controlled through norm constraints, tree depth, early stopping, margin, or algorithmic stability.

More precisely, let Y^{new} be an independent response vector at the same fixed design points and define $R_{\text{train}} = n^{-1} \left\| Y - \widehat{Y} \right\|^2$. Then

$$\frac{1}{n}\mathbb{E} \left\| Y^{\text{new}} - \widehat{Y} \right\|^2 = \mathbb{E}(R_{\text{train}}) + \frac{2\sigma^2}{n} \operatorname{df}(\widehat{Y}). \tag{0.7.2}$$

For a fixed d-parameter least-squares model, this suggests the corrected error estimate

$$\widehat{R}_{C_p} = \frac{\text{RSS}}{n} + \frac{2\widehat{\sigma}^2 d}{n},$$

or equivalently the classical standardized form $C_p = \text{RSS}\,/\widehat{\sigma}^2 - n + 2d$. If variable selection is adaptive, substituting only the selected coefficient count for d ignores search degrees of freedom and can under-penalize the procedure.

0.8 Ridge Regression

Ridge regression solves

$$\widehat{\beta}_\lambda = \arg\min_\beta \{ \|y - X\beta\|^2 + \lambda \|\beta\|^2 \}, \qquad \lambda \geq 0, \tag{0.8.1}$$

giving

$$\widehat{\beta}_\lambda = (X^\mathsf{T} X + \lambda I)^{-1} X^\mathsf{T} y.$$

For centered X and y, differentiating the objective gives

$$\nabla_\beta \{ \|y - X\beta\|^2 + \lambda \|\beta\|^2 \} = -2X^\mathsf{T} y + 2(X^\mathsf{T} X + \lambda I)\beta,$$

so the ridge normal equations are $(X^\mathsf{T} X + \lambda I)\widehat{\beta}_\lambda = X^\mathsf{T} y$. For every nonzero v and $\lambda > 0$,

$$v^\mathsf{T}(X^\mathsf{T} X + \lambda I)v = \|Xv\|^2 + \lambda \|v\|^2 > 0.$$

Thus $X^\mathsf{T} X + \lambda I$ is positive definite and the ridge solution is unique even when predictors are linearly dependent or $p > n$.

The intercept is normally left unpenalized. Predictors must be standardized because the penalty depends on units. Ridge is also the posterior mode and mean under a Gaussian prior in the Gaussian linear model.

Concretely, a common transformation is

$$Z_{ij} = \frac{X_{ij} - \bar{X}_j}{s_j}, \qquad s_j^2 = \frac{1}{n}\sum_{i=1}^n (X_{ij} - \bar{X}_j)^2.$$

Without it, changing a predictor from dollars to thousands of dollars changes the corresponding squared-coefficient penalty by a factor of 10^6. Centering and scaling constants must be estimated within each CV training fold and then applied unchanged to its validation fold.

0.8.1 Shrinkage

Conditional on X under $y = X\beta + \varepsilon$,

$$\mathbb{E}[\widehat{\beta}_\lambda] = (X^\mathsf{T}X + \lambda I)^{-1}X^\mathsf{T}X\beta,$$

so ridge is biased. It can nevertheless lower prediction MSE by greatly reducing variance, especially under collinearity or $p \geq n$. Unlike lasso, ridge usually retains every feature.

As λ increases, coefficient norm and effective degrees of freedom decrease. Tune λ by validation, fitting standardization within each fold. The constrained form $\|\beta\|^2 \leq t$ is equivalent for a matching Lagrange multiplier.

0.8.2 SVD

Let the centered design have thin SVD $X = UDV^\mathsf{T}$, with singular values $d_1 \geq \cdots \geq d_r > 0$. Then

$$\widehat{\beta}_\lambda = V \operatorname{diag}\left\{\frac{d_j}{d_j^2 + \lambda}\right\} U^\mathsf{T}y, \quad \widehat{y}_\lambda = U \operatorname{diag}\left\{\frac{d_j^2}{d_j^2 + \lambda}\right\} U^\mathsf{T}y.$$

Ridge shrinks low-variance directions most strongly. Its effective degrees of freedom are

$$\mathrm{df}(\lambda) = \sum_{j=1}^{r} \frac{d_j^2}{d_j^2 + \lambda}.$$

This counts penalized slope complexity; add one when an intercept is fit but left unpenalized.

The SVD also exposes what ridge changes relative to OLS. When $r < p$, every OLS solution has the form

$$\widehat{\beta}_{\mathrm{OLS}} = VD^{-1}U^\mathsf{T}y + w, \qquad V^\mathsf{T}w = 0.$$

The two terms are orthogonal, so $w = 0$ gives the unique minimum-norm OLS solution. In singular directions,

$$\widehat{\beta}_{\mathrm{OLS}}^{\min} = \sum_{j=1}^{r} \frac{u_j^\mathsf{T}y}{d_j} v_j,$$
$$\widehat{\beta}_\lambda = \sum_{j=1}^{r} \frac{d_j}{d_j^2 + \lambda}(u_j^\mathsf{T}y)v_j.$$

Hence ridge multiplies the OLS coefficient in direction v_j by $d_j^2/(d_j^2 + \lambda)$, shrinking weakly identified directions with small d_j most strongly. The SVD expression explains stability, shrinkage, and why ridge remains well-defined when $X^\mathsf{T}X$ is singular.

0.8.3 Bias–Variance Tradeoff

Write the fixed-design model as $y = \mu + \varepsilon$ with $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$, and define the ridge smoother

$$H_\lambda = X(X^\mathsf{T}X + \lambda I)^{-1}X^\mathsf{T} = U \operatorname{diag}\left\{\frac{d_j^2}{d_j^2 + \lambda}\right\} U^\mathsf{T}.$$

Then

$$\mathrm{Bias}(\widehat{y}_\lambda \mid X) = (H_\lambda - I)\mu, \qquad \mathrm{Var}(\widehat{y}_\lambda \mid X) = \sigma^2 H_\lambda^2.$$

Consequently the average variance contribution to prediction error is

$$\frac{1}{n}\,\mathrm{tr}\{\mathrm{Var}(\widehat{y}_\lambda \mid X)\} = \frac{\sigma^2}{n}\sum_{j=1}^{r}\left(\frac{d_j^2}{d_j^2+\lambda}\right)^2. \tag{0.8.2}$$

Increasing λ decreases this variance monotonically while generally increasing squared bias. Cross-validation estimates the value of λ that best balances the two for prediction; the one-standard-error rule chooses the largest penalty whose CV error is within one standard error of the minimum.

0.9 Lasso Regression

The lasso solves

$$\widehat{\beta}_\lambda = \arg\min_\beta \left\{\frac{1}{2}\left\|y - X\beta\right\|^2 + \lambda\left\|\beta\right\|_1\right\}. \tag{0.9.1}$$

It is convex but nonsmooth at zero. The ℓ_1 constraint has corners, so solutions often lie on coordinate axes and are sparse.

0.9.1 Sparsity

Lasso performs shrinkage and variable selection together. With correlated predictors it may select one arbitrarily, making the selected set unstable. Ridge tends to distribute weight across the group. Elastic net combines ℓ_1 and squared ℓ_2 penalties and often stabilizes grouped features.

Subset selection can be written as an ℓ_0-penalized problem,

$$\min_\beta \left\{\frac{1}{2}\left\|y - X\beta\right\|^2 + \lambda_0\left\|\beta\right\|_0\right\}, \qquad \left\|\beta\right\|_0 = \sum_{j=1}^{p}\mathbb{1}\{\beta_j \neq 0\}.$$

It charges the same amount for every retained predictor, regardless of its coefficient magnitude. This objective is nonconvex and generally requires a combinatorial search because choices interact when predictors are correlated. The lasso replaces the discontinuous ℓ_0 count by the convex relaxation $\left\|\beta\right\|_1$. Geometrically, an ℓ_2 constraint is a smooth ball, whereas the ℓ_1 ball has corners on coordinate axes; an RSS contour often first touches an ℓ_1 corner, producing exact zeros.

0.9.2 Orthonormal Design: Hard, Soft, and Ridge Shrinkage

Suppose the response and predictors are centered and $X^\mathsf{T}X = I$. Define $z = X^\mathsf{T}y = \widehat{\beta}_{\mathrm{OLS}}$. Then

$$\frac{1}{2}\left\|y - X\beta\right\|^2 = \text{constant} + \frac{1}{2}\sum_{j=1}^{p}(\beta_j - z_j)^2,$$

so each penalized problem separates coordinate by coordinate.

For best subset selection, coordinate j compares the cost $z_j^2/2$ from setting $b = 0$ with the cost λ_0 from setting $b = z_j$. Therefore

$$\widehat{\beta}_j^{\ell_0} = z_j \mathbb{1}\{|z_j| > \sqrt{2\lambda_0}\}, \tag{0.9.2}$$

which is *hard thresholding*: retain the OLS coefficient unchanged or delete it.

For the lasso, minimize $q(b) = \frac{1}{2}(b - z_j)^2 + \lambda_1|b|$. Away from zero,

$$q'(b) = \begin{cases} b - z_j + \lambda_1, & b > 0, \\ b - z_j - \lambda_1, & b < 0. \end{cases}$$

At zero the subgradient condition is $0 \in -z_j + \lambda_1[-1, 1]$, which holds exactly when $|z_j| \le \lambda_1$. Hence

$$\widehat{\beta}_j^{\text{lasso}} = \text{sign}(z_j)(|z_j| - \lambda_1)_+, \tag{0.9.3}$$

the soft-thresholding rule. In contrast, ridge with objective $\frac{1}{2}\|y - X\beta\|^2 + \frac{\lambda_2}{2}\|\beta\|^2$ gives

$$\widehat{\beta}_j^{\text{ridge}} = \frac{z_j}{1 + \lambda_2}. \tag{0.9.4}$$

Thus ℓ_0 deletes or retains, lasso deletes or shrinks, and ridge shrinks every coordinate continuously.

Exact support recovery needs conditions stronger than those needed for good prediction, including sufficient signal and restrictions on correlations. A sparse fitted model should not automatically be interpreted as identifying the true causal variables.

The preferred penalty depends on signal structure. Lasso is often effective when relatively few predictors carry signal, because setting irrelevant coefficients to zero can substantially reduce variance. Ridge can perform better when signal is dense or distributed across correlated groups, because deleting coefficients creates unnecessary bias and lasso may select one member of a correlated group arbitrarily. One-hot encoded levels and basis expansions are examples where group-aware penalties may be preferable to coordinatewise lasso.

0.9.3 KKT

Let $r = y - X\widehat{\beta}$. The Karush–Kuhn–Tucker conditions are

$$X_j^\mathsf{T} r = \begin{cases} \lambda \,\text{sign}(\widehat{\beta}_j), & \widehat{\beta}_j \neq 0, \\ z_j, \quad |z_j| \le \lambda, & \widehat{\beta}_j = 0. \end{cases} \tag{0.9.5}$$

Thus a variable stays at zero when its correlation with the current residual does not exceed λ. These conditions certify optimality and support screening rules.

If $\lambda \ge \left\|X^\mathsf{T} y\right\|_\infty$, the all-zero vector is optimal when the response and columns have been centered.

0.9.4 Coordinate Descent

Holding all other coefficients fixed, define the partial residual $r_j = y - \sum_{k \neq j} X_k \beta_k$. The coordinate update is

$$\beta_j \leftarrow \frac{S(X_j^\top r_j, \lambda)}{\|X_j\|^2}, \qquad S(z, \lambda) = \operatorname{sign}(z)(|z| - \lambda)_+. \tag{0.9.6}$$

The soft-thresholding operator sets small correlations exactly to zero. Cyclic coordinate descent is efficient for sparse matrices. Warm starts along a decreasing grid of λ values greatly reduce computation; convergence should be checked through KKT residuals or the primal–dual gap.

0.10 Regression Splines and Basis Expansions

Regularization reduces estimation variance within a chosen model class, but it cannot remove approximation bias caused by an overly restrictive class. For example, shrinking the coefficients of $\beta_0 + \beta_1 x$ still leaves a straight line. A basis expansion instead writes

$$f(x) = \sum_{j=0}^{M} \beta_j h_j(x).$$

For a fixed collection of basis functions, define $B_{ij} = h_j(x_i)$. Then fitting remains ordinary linear regression:

$$\widehat{\beta} = (B^\top B)^{-1} B^\top y$$

when B has full column rank. The fitting algorithm is unchanged; what has changed is the linear space of candidate functions.

0.10.1 Polynomial and Local Bases

Polynomial regression uses $h_j(x) = x^j$:

$$f(x) = \beta_0 + \beta_1 x + \cdots + \beta_d x^d.$$

It is nonlinear in x but linear in its coefficients, and degree d can be chosen by CV. Increasing d lowers approximation bias but raises estimation variance and can make the fit unstable near boundaries. With p predictors, a complete polynomial expansion through total degree d contains

$$M(p, d) = \binom{p + d}{d}$$

terms including the intercept. Thus explicitly enumerating nonlinearities and interactions quickly becomes combinatorial.

Step functions provide local constant fits by using interval indicators. They are easy to interpret but discontinuous. A more useful local construction uses the hinge function

$$(x - \xi)_+ = \max(x - \xi, 0).$$

A linear spline with interior knots $\xi_1, \ldots, \xi_K$ has representation

$$f(x) = \beta_0 + \beta_1 x + \sum_{k=1}^{K} \theta_k (x - \xi_k)_+. \tag{0.10.1}$$

It is continuous, has $K + 2$ coefficients, and its slope changes by θ_k after knot ξ_k.

0.10.2 Cubic Regression Splines

A degree-d regression spline is piecewise polynomial with continuity constraints at knots $\xi_1, \ldots, \xi_K$. A cubic spline requires the function and its first two derivatives to agree across every knot. Counting parameters makes its dimension transparent: $K + 1$ cubic pieces have $4(K + 1)$ raw coefficients, while value, slope, and curvature matching impose $3K$ constraints. Hence the spline space has dimension

$$4(K + 1) - 3K = K + 4.$$

The truncated-power basis

$$1, x, x^2, x^3, (x - \xi_1)^3_+, \ldots, (x - \xi_K)^3_+$$

has exactly this size and automatically enforces the smoothness constraints. More generally, a degree-d truncated-power basis is

$$1, x, \ldots, x^d, (x - \xi_1)^d_+, \ldots, (x - \xi_K)^d_+.$$

B-spline bases span the same spline space but have local support and much better numerical conditioning. Each cubic B-spline basis function is nonzero on at most four consecutive knot spans, so changing one coefficient while holding the others fixed changes the represented function only locally. This does *not* mean that changing one observation and refitting has only a local effect: least squares may change many coefficients, so the fitted curve can move throughout the domain.

With a fixed basis matrix B, fitting is linear regression. Flexibility is controlled by degree, knot locations, number of knots, or a roughness penalty. Knots are often placed at empirical quantiles so data-rich regions receive more resolution, while CV can select their number. Boundary behavior matters: natural cubic splines impose $f'' = 0$ at both boundary knots and extend linearly outside them. With K interior knots their dimension is $K + 2$, two fewer than an unrestricted cubic spline, and their tail behavior is typically more stable than that of a high-degree global polynomial.

Local support alone does not solve high-dimensional feature growth. If each of p predictors has m basis functions, a full tensor-product basis has m^p terms. Additive models avoid this explosion by excluding interactions unless they are introduced deliberately.

0.10.3 Smoothing Splines

The cubic smoothing spline minimizes

$$\sum_{i=1}^{n}\{y_i - f(x_i)\}^2 + \lambda \int \{f''(t)\}^2 \, \mathrm{d}t \tag{0.10.2}$$

over functions with square-integrable second derivative. The minimizer is a natural cubic spline with knots at the distinct observed x_i values. Thus an apparently infinite-dimensional optimization has a finite-dimensional solution.

For a suitable basis, the coefficient problem is

$$\widehat{\beta} = (B^{\mathsf{T}}B + \lambda\Omega)^{-1}B^{\mathsf{T}}y, \qquad \Omega_{jk} = \int B_j''(t)B_k''(t)\,\mathrm{d}t.$$

The fitted values are $S_\lambda y$ and $\mathrm{df}(\lambda) = \mathrm{tr}(S_\lambda)$. As $\lambda \to 0$ the fit interpolates; as $\lambda \to \infty$ it approaches a straight line. Generalized cross-validation approximates leave-one-out error by

$$\mathrm{GCV}(\lambda) = \frac{n^{-1}\,\|(I - S_\lambda)y\|^2}{\{1 - \mathrm{tr}(S_\lambda)/n\}^2}.$$

0.10.4 GAMs

A generalized additive model (GAM) assumes

$$g\{\mathbb{E}(Y \mid X = x)\} = \beta_0 + \sum_{j=1}^{p} f_j(x_j),$$

where g is a link function. Each component is represented by a basis and regularized with a smoothness penalty. Identifiability commonly uses $\sum_i f_j(x_{ij}) = 0$.

GAMs capture nonlinear main effects while retaining component plots and efficient fitting. Backfitting alternately updates each f_j using partial residuals. Penalized IRLS handles non-Gaussian responses. The additive assumption excludes interactions unless terms such as tensor-product smooths are explicitly included.

0.11 Decision Trees

A regression tree partitions feature space into terminal regions $R_1, \dots, R_M$ and predicts

$$\widehat{f}(x) = \sum_{m=1}^{M} \widehat{c}_m \mathbb{1}\{x \in R_m\}, \qquad \widehat{c}_m = \frac{1}{|R_m|}\sum_{i : x_i \in R_m} y_i.$$

At each node, CART greedily chooses a feature j and threshold s minimizing the sum of squared errors in the two child nodes. For classification, common node impurities are

$$Q_{\mathrm{Gini}} = 1 - \sum_k \widehat{p}_k^2, \qquad Q_{\mathrm{entropy}} = -\sum_k \widehat{p}_k \log \widehat{p}_k.$$

Trees model interactions automatically, need little preprocessing, and are invariant to monotone transformations of individual features. A single tree is unstable: small data changes can alter early splits and the whole partition. Greedy training is computationally feasible but does not find the globally optimal tree.

Continuous split search can be done after sorting each feature. Missing values require an explicit strategy, such as imputation, learned default directions, or surrogate splits. Unrestricted high-cardinality categorical features can receive an unfairly large chance to reduce training impurity.

0.11.1 Pruning

Grow a sufficiently large tree T_0, then choose a subtree by cost-complexity pruning:

$$C_\alpha(T) = \sum_{m=1}^{|T|} \sum_{i:x_i \in R_m} (y_i - \widehat{c}_m)^2 + \alpha\,|T|\,. \tag{0.11.1}$$

Weakest-link pruning produces a nested sequence of optimal subtrees as α increases. Cross-validation chooses α; the one-standard-error rule often yields a smaller, more stable tree.

Stopping growth early can miss a useful split whose benefit appears only after another split. Growing then pruning separates search from complexity selection.

0.12 Random Forests

Random forests average many deep randomized trees:

$$\widehat{f}_{\mathrm{RF}}(x) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}_b(x).$$

Each tree uses a bootstrap sample, and each split considers a random subset of features. Feature subsampling decorrelates trees; averaging then reduces variance more effectively.

If identically distributed tree predictions have variance σ^2 and pairwise correlation ρ, the average of B trees has variance

$$\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Increasing B removes only the second term. Forest performance therefore depends on both individual tree strength and inter-tree correlation.

Out-of-bag predictions evaluate observation i using only trees whose bootstrap samples excluded it. They provide a convenient internal error estimate, though an external test set is still needed after extensive tuning.

0.12.1 Bagging

Bagging fits the same unstable learner to bootstrap samples and averages:

$$\widehat{f}_{\mathrm{bag}}(x) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}^{*(b)}(x).$$

For squared loss, averaging reduces variance. It helps trees because they are unstable, but changes little for an already stable linear estimator.

Permutation importance measures the loss increase after shuffling a feature in out-of-bag data. It measures predictive reliance, not causal effect, and can split importance among correlated variables. Impurity-based importance is biased toward continuous or high-cardinality features.

0.13 Boosting as Functional Gradient Descent

Boosting builds an additive model

$$F_M(x) = F_0(x) + \sum_{m=1}^{M} \nu\gamma_m h_m(x)$$

by sequentially fitting weak learners. For differentiable empirical loss $\sum_i L(y_i, F(x_i))$, iteration m computes pseudo-residuals

$$r_{im} = -\left.\frac{\partial L(y_i, F)}{\partial F}\right|_{F=F_{m-1}(x_i)},$$

fits h_m to (x_i, r_{im}), chooses a step γ_m, and updates $F_m = F_{m-1} + \nu\gamma_m h_m$. Thus boosting is gradient descent in function space.

For squared loss, pseudo-residuals are ordinary residuals. For binary logistic loss they are closely related to $y_i - \widehat{p}_i$. Tree depth controls interaction order; learning rate ν and number of trees trade off. Smaller ν usually requires more iterations and often generalizes better.

Boosting can overfit, especially with deep trees, noisy labels, or excessive iterations. Tune on validation data and monitor training and validation loss. Unlike bagging, trees are dependent and sequential: each corrects errors of the current ensemble.

Algorithm 1 Gradient boosting

1: Initialize $F_0(x) = \arg\min_c \sum_i L(y_i, c)$.
2: **for** $m = 1, \ldots, M$ **do**
3: Compute $r_{im} = -\partial L(y_i, F)/\partial F$ at $F = F_{m-1}(x_i)$.
4: Fit a weak learner h_m to $\{(x_i, r_{im})\}$.
5: Choose γ_m by line search.
6: Set $F_m = F_{m-1} + \nu\gamma_m h_m$.
7: **end for**

0.14 Maximum Margin Classifiers

For labels $y_i \in \{-1, 1\}$, a linear classifier predicts $\text{sign}(\beta_0 + x^\mathsf{T}\beta)$. For separable data, the hard-margin SVM minimizes $\|\beta\|^2/2$ subject to $y_i(\beta_0 + x_i^\mathsf{T}\beta) \geq 1$. Its geometric margin is $1/\|\beta\|$. Feature scaling matters because distance and the penalty depend on units.

0.14.1 Soft-Margin SVM

Nonseparable data use slack variables:

$$\min_{\beta_0,\beta,\xi} \frac{1}{2}\|\beta\|^2 + C\sum_i \xi_i \quad \text{subject to} \quad y_i(\beta_0 + x_i^\mathsf{T}\beta) \geq 1 - \xi_i, \quad \xi_i \geq 0. \tag{0.14.1}$$

Equivalently, minimize $\|\beta\|^2/2 + C\sum_i [1 - y_i(\beta_0 + x_i^\mathsf{T}\beta)]_+$. Large C gives less regularization; small C permits violations for a wider margin.

0.14.2 SVM Duality

Eliminating the primal variables gives

$$\max_\alpha \left\{ \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i\alpha_j y_i y_j x_i^\mathsf{T} x_j \right\} \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0. \tag{0.14.2}$$

The primal coefficient is $\widehat{\beta} = \sum_i \widehat{\alpha}_i y_i x_i$. Convexity and feasibility give strong duality.

0.14.3 Support Vectors and Kernels

KKT complementary slackness shows that observations with $\alpha_i = 0$ do not directly affect the boundary. Those with $\alpha_i > 0$ are support vectors: $0 < \alpha_i < C$ indicates a point on the margin, while $\alpha_i = C$ indicates a point inside the margin or misclassified.

A positive-semidefinite kernel $K(x, z) = \langle \phi(x), \phi(z) \rangle$ implicitly maps inputs to a feature space. Replacing inner products in (0.14.2) gives

$$f(x) = \widehat{\beta}_0 + \sum_i \widehat{\alpha}_i y_i K(x_i, x).$$

Polynomial and Gaussian radial kernels are common. Kernel width and C must be tuned jointly. The Gram matrix costs $O(n^2)$ memory. The raw margin is not a probability; calibrate it on held-out data if probabilities are required.

0.15 PCA from Constrained Optimization

For centered $X \in \mathbb{R}^{n\times p}$, the first principal loading solves

$$v_1 = \arg\max_{\|v\|=1} \text{Var}(Xv) = \arg\max_{\|v\|=1} v^\mathsf{T} X^\mathsf{T} X v. \tag{0.15.1}$$

The Lagrange equation makes v_1 the leading covariance eigenvector. Subsequent loadings maximize variance subject to orthogonality. If $X = UDV^{\mathsf{T}}$, scores are UD, loadings are V, and component j explains variance proportional to d_j^2. Standardize columns when their scales are not comparable. PCA is unsupervised: high variance need not predict Y.

0.15.1 PCA as LoRA

Here LoRA means low-rank approximation. The Eckart–Young–Mirsky theorem states

$$X_k = U_k D_k V_k^{\mathsf{T}} = \underset{\operatorname{rank}(A)\le k}{\arg\min} \|X - A\|_F^2, \qquad \|X - X_k\|_F^2 = \sum_{j>k} d_j^2. \tag{0.15.2}$$

Modern low-rank adaptation uses an update $\Delta W = AB^{\mathsf{T}}$. It shares the representation idea, but its factors optimize a supervised objective rather than arise from a data SVD.

0.15.2 PCR connection

Principal components regression regresses y on the first k scores:

$$\widehat{\beta}_{\mathrm{PCR},k} = V_k D_k^{-1} U_k^{\mathsf{T}} y.$$

PCR discards low-variance directions; ridge shrinks them continuously. Because PCR selects directions without y, a low-variance component can be highly predictive. Select k by CV rather than explained variance alone.

0.16 K-means

K-means minimizes

$$\min_{C_1,\dots,C_K,\,\mu_1,\dots,\mu_K} \sum_{k=1}^{K} \sum_{i\in C_k} \|x_i - \mu_k\|^2. \tag{0.16.1}$$

Lloyd's algorithm alternates nearest-center assignment with updating each center to its cluster mean. The objective decreases, but convergence is only to a local optimum. Use K-means++ and multiple initializations, standardize differently scaled features, and handle empty clusters explicitly.

K-means favors roughly spherical, similarly sized clusters and is sensitive to outliers. Choose K using stability, substantive usefulness, silhouette summaries, or a model-based criterion; no single diagnostic defines the scientifically correct number.

0.17 Hierarchical Clustering

Agglomerative clustering starts with singleton clusters and repeatedly merges the closest pair. Common linkages are

$$d_{\text{single}}(A,B) = \min_{i\in A, j\in B} d(i,j),$$
$$d_{\text{complete}}(A,B) = \max_{i\in A, j\in B} d(i,j),$$
$$d_{\text{average}}(A,B) = \frac{1}{|A|\,|B|} \sum_{i\in A, j\in B} d(i,j).$$

Ward linkage chooses the smallest increase in within-cluster sum of squares.

The dendrogram records merge heights; branch order within a merge is not meaningful. Single linkage can recover elongated clusters but chains through nearby points. Complete linkage favors compact clusters. Distance, scaling, and preprocessing define similarity. A dendrogram always displays structure, even for homogeneous noise, so assess stability by perturbation or bootstrap co-clustering and require substantive validation.